Seasonal Variation of Salinity in a Small Tidal Estuary

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Abstract

We report twelve months of salinity measurements from a small tidal estuary. Salinity at the upper station fell sharply during the wet season and recovered within six weeks. The results suggest that river discharge, rather than tidal range, controls the position of the salt front.

1 Introduction

Small estuaries respond quickly to changes in river flow. Few long records exist for estuaries of this size in the region. Most published studies cover large systems with deep, dredged channels. This paper describes a full year of observations and compares them with the river gauge record.

2 Methods

Conductivity loggers were installed at three stations along the channel. Each logger recorded a value every fifteen minutes. The loggers were cleaned and checked against a hand-held meter once a month. River discharge was taken from a gauge located four kilometres upstream.

3 Results

The lower station showed little seasonal change. At the upper station, mean salinity dropped from 18 to 3 practical salinity units in January. The middle station followed the upper station with a delay of about ten days. Spring tides raised salinity at all stations, but the effect lasted less than two days. No relation was found between wind direction and the position of the salt front.

4 Conclusion

River discharge is the dominant control on salinity in this estuary. Future work will extend the record to a second estuary with a larger tidal range.